

1 Context

In this deliverable, you will study a spatial search problem inspired by the recovery of a lost object at sea after an aerial accident. The search region is represented by a fixed rectangular grid. The true location of the object is unknown, and search missions provide only imperfect information.

The goal of the assignment is to construct a Bayesian model on this grid. In particular, you will:

- build a **prior distribution** over the cells of the grid,
- define a **detection model** for search missions,
- **update the prior** using search outcomes,
- use the posterior distribution to **propose a new search mission**.

The assignment is intentionally designed so that some ingredients are fixed and common to all students, while others must be modeled and justified by you.

2 Provided data

You will have access to the dataset `grid_dataset`, which describes the fixed grid used in the problem. Each row corresponds to one valid cell. The dataset contains the following columns:

Variable	Description
<code>cell_id</code>	unique identifier of the cell
<code>x</code>	x -coordinate of the center of the cell
<code>y</code>	y -coordinate of the center of the cell
<code>depth</code>	normalized depth associated with the cell
<code>roughness</code>	normalized seafloor roughness associated with the cell

The dimensions of the grid are fixed and common to all students: the number of columns is $N_x = 50$ and the number of rows is $N_y = 35$. In addition, you will be given the following information about the accident in notebook format:

- the explosion point of the aircraft, expressed in grid coordinates,
- the aircraft velocity vector, expressed in grid units,

- a wind vector,
- a water drift/current vector,
- two witness statements.

You may use all, part, or none of this information in your modeling choices, but **any choice must be justified**.

3 Fixed rules of the problem

Some elements of the problem are fixed and must be used exactly as stated. In particular, the spatial discretization is not part of the modeling task: all students must work on the same grid and on the same cell-level dataset.

A **search mission** t is defined by a searched region $R_t \subseteq \mathcal{G}$, where $\mathcal{G}$ denotes the set of valid cells of the grid, together with an effort level e_t . Once a searched region has been selected, the coverage coefficient c_{tj} for mission t and cell j is determined by

$$c_{tj} = \begin{cases} 1, & \text{if } j \in R_t, \\ 0, & \text{if } j \notin R_t, \end{cases} \quad (1)$$

and the cost of mission t is defined by

$$\text{cost}_t = e_t |R_t|, \quad (2)$$

where $|R_t|$ denotes the number of cells in the searched region. A total budget B is imposed, so any admissible sequence of search missions must satisfy

$$\sum_{t=1}^T e_t |R_t| \leq B.$$

Finally, if the object is located in cell j , the detection probability of mission t must be modeled in the form

$$q_{tj} = c_{tj} \rho_{tj}, \quad (3)$$

where ρ_{tj} denotes the conditional probability of detection given that the cell is effectively covered. The functional form of ρ_{tj} is not fixed in advance: proposing, justifying, and implementing a reasonable model for this quantity is part of the assignment.

4 Open modeling decisions

The following components must be proposed and properly justified by you.

Prior distribution

You must define a prior distribution over the cells of the grid:

$$\pi_j = P(Z = j) \quad \forall j \in \mathcal{G},$$

where Z denotes the unknown true cell containing the object. Moreover, your prior must satisfy

$$\pi_j \geq 0 \quad \forall j \in \mathcal{G}, \quad \text{and} \quad \sum_{j \in \mathcal{G}} \pi_j = 1.$$

The prior should be constructed from the information provided about the accident. You are free to decide whether all of these ingredients are useful in your model. However, your choice must be explained and justified.

Detection model

You must define a model for ρ_{tj} , the conditional probability of detection given coverage. You can gain further insight into the problem by reading the PDF file `previous_missions_reports.pdf`. You may use all, part, or none of this information, but your model must produce valid probabilities and must be justified.

Mission reports from previous search attempts carried out in similar environments are not part of the formal dataset, but they may contain useful qualitative information about detectability. You may use this information to motivate the form of your detection model or to justify the expected qualitative effect of some variables.

5 Search missions

You will carry out search missions through the following webpage: <https://search-missions.streamlit.app/>. For each mission, you must specify a rectangular searched region by providing the bounds $x_{\min}, x_{\max}, y_{\min}, y_{\max}$, together with an effort level e_t . The webpage will then return

$$s_t \in \{0, 1\},$$

where $s_t = 1$ means detection and $s_t = 0$ means no detection.

These observed missions must then be used to update your prior and construct a posterior distribution over the grid. For the submission, you must also provide the file `missions.csv`, containing the history of your search missions together with their observed outcomes. The file must include the following fields:

```
team_id, mission_id, x_min, x_max, y_min, y_max, effort, n_cells, cost, s_t, budget_remaining, timestamp
```

6 Tasks – Brief Summary

In summary, your work must address the following tasks.

Task 1: Prior construction

Construct a prior distribution $\{\pi_j\}_{j \in \mathcal{G}}$ over the grid cells. You must:

- explain how the provided physical and testimonial information is used,
- justify the modeling choices made,
- show a heatmap of the resulting prior distribution.

Task 2: Detection model

Define a model for ρ_{tj} and therefore for

$$q_{tj} = c_{tj}\rho_{tj}.$$

You must:

- explain the variables included in the model,
- justify the expected qualitative effect of each variable,
- ensure that the model returns valid probabilities.

Task 3: Posterior update

Using the observed search missions, update the prior and compute a posterior distribution over the grid. You must:

- derive the likelihood associated with your model,
- combine prior and likelihood to obtain the posterior,
- show a heatmap of the posterior distribution.

Task 4: Next search mission

Using your posterior distribution, propose one additional search mission. Your proposal must:

- specify a searched region R_{T+1} ,
- specify an effort level e_{T+1} ,
- respect the budget constraint,
- be justified using your current posterior beliefs.

Task 5: Discussion

Briefly discuss the main strengths of your model, the main limitations of your assumptions, and which elements of the model you would improve if more information were available.

7 Submission format

The submission must contain two separate parts. First, you must submit a short written report of at most two pages. This report should summarize only the most relevant aspects of your work, such as how the prior distribution is constructed, which model is used for the detection probability, and how the proposed new search missions are selected and justified.

Second, you must submit a summary notebook containing the main figures, the essential code, and the basic elements needed to review the modeling and updating process. The notebook should make the workflow clear and reproducible, but it is not expected to include every intermediate exploration carried out during development. All figures and numerical results must be generated from the provided dataset and the code you submit.

8 Final remarks

This deliverable is designed so that there is no single correct model. Different choices may be reasonable, provided they are mathematically coherent, well justified, and correctly implemented. The emphasis is placed not only on the final answer, but also on the quality of the probabilistic reasoning used to construct the model.